

EDUCATION

University of Southern California

Ph.D. in Computer Science; GPA: 4.0

Los Angeles, CA

Aug 2020 – May 2024

Dissertation: Dispersed Computing in Dynamic Environments

Advisor: Bhaskar Krishnamachari

California State University, Long Beach

Master of Science in Computer Science; GPA: 4.0

Long Beach, CA

Aug 2018 – May 2020

Bachelor of Science in Computer Science; GPA: 3.6

Aug 2013 – May 2018

Thesis: Minimizing The Maximum Distance Traveled To Form Patterns With Systems of Mobile Robots

Advisor: Oscar Morales-Ponce

RESEARCH INTERESTS

Cooperative multi-agent systems, distributed computing, decentralized computing, online algorithms, distributed ledger technology, internet of things (IoT), artificial intelligence (AI), and large language models (LLMs) for endangered language revitalization.

EXPERIENCE

University of Southern California

Adjunct Research Assistant Professor of Electrical and Computer Engineering

Los Angeles, CA

July 2024 – Present

Coinbase

Consultant - Machine Learning Engineer

Remote

Jan 2024 – Present

Loyola Marymount University

Assistant Professor of Computer Science

Los Angeles, CA

Aug 2024 – Present

The Aerospace Corporation

Member of the Technical Staff

El Segundo, CA

Mar 2020 – Jun 2024

Associate Member of the Technical Staff

Sep 2018 – Mar 2020

Intern

Jan 2018 – Aug 2018

- Designed software and algorithms for rapidly developing scalable, modular, and efficient analyses for launch vehicle verification in simulation, day-of-launch, and post-flight environments.

CSULB Research Foundation

Student Research Assistant

Long Beach, CA

Mar 2017 – May 2018

- Developed software and simulations for systems of cooperative robots.

Big Pine Paiute Tribe of the Owens Valley

IT Intern

Big Pine, CA

May 2014 – Aug 2014

- Organized computer skills workshops for high school students in a tribal work experience program
- Developed membership management software for the tribal wellness center

TEACHING EXPERIENCE

Online and Decentralized Algorithms

Professor

Loyola Marymount University

Spring 2025

Algorithms and Analysis

Professor

Loyola Marymount University

Fall 2024, Spring 2025, Fall 2025

PROFESSIONAL SERVICE

ACM/IEEE Symposium on Edge Computing <i>Reviewer</i>	<i>June, 2026</i>
Swiss National Science Foundation <i>Proposal Reviewer</i>	<i>April, 2026</i>
IEEE Transactions on Sensor Networks <i>Reviewer</i>	<i>April, 2026</i>
NetCompute 2026 — 1st Workshop on Computation over Heterogeneous Networks @ INFOCOM <i>Workshop Chair / Organizer</i>	<i>May, 2026</i>
Indigenous Working Group @ LMU <i>Chair (elected October 2025); Member since 2024</i>	<i>2024–present</i>
Workshop on AI and Indigenous Languages <i>Organizing Committee Chair</i>	<i>September, 2025–present</i>
Seaver College DEI Committee <i>Member</i>	<i>2025–present</i>
LMU GenAI Task Force <i>Member</i>	<i>August, 2025–present</i>
CS GenAI Task Force @ LMU <i>Member</i>	<i>July, 2025–present</i>
Theoretical Computer Science Journal <i>Reviewer</i>	<i>December, 2025</i>
Transactions on Computers <i>Reviewer</i>	<i>September, 2025</i>
AAAI 2026 — Special Track on AI for Social Impact <i>Program Committee Member</i>	<i>September, 2025</i>
WISDOM 2025 Workshop <i>Program Committee Member</i>	<i>San Diego, CA July, 2025</i>
Discrete Applied Mathematics <i>Reviewer</i>	<i>June, 2025</i>
National Science Foundation <i>DLI-DEL Advisory Panel Member</i>	<i>Spring 2025</i>
CCGrid 2025 — Symposium on Cluster, Cloud, and Internet Computing <i>Artifact Track Program Committee Member</i>	<i>Tromso, Norway February, 2025</i>
SC25 — International Conference for HPC, Networking, Storage, and Analysis <i>Reproducibility Committee Member</i>	<i>St. Louis, MO February, 2025</i>
International Journal of Computing and Technology <i>Reviewer</i>	<i>January, 2025</i>
Journal of Computer and System Sciences <i>Reviewer</i>	<i>December, 2024</i>
GHTC 2024 — IEEE Global Humanitarian Technology Conference <i>Track Chair</i>	<i>Philadelphia, PA Oct 23-26, 2024</i>
SC24 — International Conference for HPC, Networking, Storage, and Analysis <i>Technical Program Committee</i>	<i>Atlanta, GA Nov 17-22, 2024</i>

MENTORSHIP / RESEARCH ADVISING

Matias Martinez Gonzalez and Nicholas Laus (Undergrad)

Task Graph Scheduling — SAGA Integrations

2025 - present

Harsh Devisha (Masters)

Cooperative Mobile Agents

2025 - present

Quinn Austin and Sebastian Lange (Undergrad)

Kubishi Scholar

2025 - present

Diego Cuadros and Nicholas Leeds (Undergrad)

Retrieval Augmented Generation for LLM-powered Low-Resource Machine Translation

2024 - present

Kathan Pathak (Masters)

Culturally Appropriate Image Generation for Language Education

2024 - 2025

Gabriel Twigg-Ho and Jason Chamorro (Undergrad)

Stochastic Task Graph Scheduling

2024 - present

Rohin Gupta (High School)

Using Simulated Annealing for Task Graph Scheduling

2024

Alex Sotiropoulos (Masters), Linus Lei (Undergrad)

Crowdsourcing LLM Fine-Tuning

2023 - 2024

Eric Cheng, Helen Hu, Max Li, Melanie Xiao Xuan Toh, Sana Jain (Undergrad)

LanguageGames: open-source modular minigames for language education

Fall 2023

Eason Qin, Flora Jia, Lorena Yan (Undergrads)

AskANRG: LLM-powered chatbots tailored for research groups

Fall 2023

Dominick Hart (High School)

LLM-powered NPCs for language education through video games

Summer 2023

Ebrahim Hirani, Ravi Vivek Agrawal (Masters), Linus Lei, Sanjana Adapala (Undergrad)

Saamarth Seth (High School)

Python implementations of task graph scheduling algorithms

2023

Chely Fernandez (Undergrad)

Blockchain-secured Industrial IoT

2022

Lorand Cheng (Undergrad)

Search and Rescue on the Line

2021

Wanwiset Peerapatanapokin (Undergrad)

Blockchain-secured Industrial IoT

2020

GRANTS & FUNDING

NSF #2542375 — Conference: Building Scientific and Collaborative Capacity for AI in Indigenous L

Principal Investigator

NSF #2451267 — CRII: CSR: RUI: Novel Approaches for Task Graph Scheduling Algorithm Design

Principal Investigator

HONORS & AWARDS

Cobell Scholar

The Cobell Scholarship

2023-2024

Awarded to “high-achieving, community-involved Native college students with leadership experience”.

CPS Rising Star

2023 CPS Rising Stars Workshop

2023

This workshop aims to identify and mentor outstanding PhD students and postdocs who are interested in pursuing academic careers in Cyber-Physical Systems (CPS) related areas

First-Place Winner

2nd Student Design Competition on Networked Computing on the Edge @ CPS IoT Week 2022

2022

Graph Convolutional Network-based Scheduler for Distributing Computation in the Internet of Robotic Things

IPDPS 2026 — 40th International Parallel and Distributed Processing Symposium

Tutorial: “Resilient Scheduling in Distributed Systems: Benchmarking and Adversarial Analysis with SAGA” May 25, 2026

ICDCN 2026 — 27th International Conference on Distributed Computing and Networking

Tutorial: “Resilient Scheduling in Distributed Systems: Benchmarking and Adversarial Analysis with SAGA” January 9, 2026

Lectures on Linguistics and Language Series, University of Illinois Chicago

Chicago, IL

Invited Speaker — LLMs and Endangered Language Revitalization

November 12, 2025

Workshop on IoT for Adversarial Environments @ MILCOM 2025

Los Angeles, CA

Invited Visionary Talk: “Rethinking Scheduling for Resilience in Mission-Critical IoT Workflows”

October 6, 2025

Institute of Mathematics and Computer Science, USP São Carlos

São Carlos, Brasil

Guest Lecture: “Problem Solving with Cooperative Mobile Agent Systems”

April 26, 2023

PUBLICATIONS

- [28] **The Power of Knowledge in Linear Search for an Escaping Target**
Jared Coleman, Dmitry Ivanov, Evangelos Kranakis, Danny Krizanc, and Oscar Morales-Ponce
Journal of Computer and System Sciences, Volume 157, 103737
- [27] **Comparing LLM-Based Translation Approaches for Extremely Low-Resource Languages**
Jared Coleman, Ruben Rosales, Kira Toal, Diego Cuadros, Nicholas Leeds, Bhaskar Krishnamachari, and Khalil Iskarous
LoResMT @ EACL 2026 — The 9th Workshop on Technologies for Machine Translation of Low Resource Languages
- [26] **Studying the Effect of Schedule Preemption on Dynamic Task Graph Scheduling**
Mohammadali Khodabandehlou, Jared Coleman, and Bhaskar Krishnamachari
To Appear at MILCOM 2025 WS-9 - Workshop On The Internet Of Things For Adversarial Environments
- [25] **Adapting Classic Scheduling Heuristics for Online Execution under Uncertainty**
Jason Chamorro, Gabriel Twigg-Ho, Jared Coleman, Tainã Coleman, Bhaskar Krishnamachari, and Mohammadali Khodabandehlou
To Appear at WORKS @ SC25 - The 20th Workshop on Workflows in Support of Large-Scale Science
- [24] **Community Voices on Digital Tools for Owens Valley Paiute**
Jared Coleman
LMU Digital Commons
- [23] **Crowd-SFT: Crowdsourcing for LLM Alignment**
Alex Sotiropoulos, Sulyab Thottungal Valapu, Linus Lei, Jared Coleman, and Bhaskar Krishnamachari
DAPPS 2025 - The 7th IEEE International Conference on Decentralized Applications and Infrastructures
- [22] **Optimal Delivery with a Faulty Drone**
Jared Coleman, Evangelos Kranakis, Danny Krizanc, and Oscar Morales-Ponce
CCCG 2025 - The 37th Canadian Conference on Computational Geometry
- [21] **Linear Search with Probabilistic Detection and Variable Speeds**
Jared Coleman, Oscar Morales-Ponce
IWOCA 2025 - The 36th International Workshop on Combinatorial Algorithms
- [20] **Extended Abstract: A Chatbot for Endangered Language Research**
Diego Cuadros, Jared Coleman, Ruben Rosales, Glenn Nelson
AmericasNLP @ NAACL 2025 - 5th Workshop on NLP for Indigenous Languages of the Americas
- [19] **Evaluating Scheduling Algorithms for Adaptive Orchestration in Federated Tactical Edge Cloud Environments**
Alessandro Amato, Harrie Bastiaansen, Jared Coleman, Willem Datema, Mattia Fogli, Johan van der Geest, Bhaskar Krishnamachari, Thomas Kudla, Pablo Sanchez, Niranjana Suri
ICMCIS 2025 — International Conference on Military Communications and Information Systems
- [18] **Multimodal Search on a Line**
Jared Coleman, Dmitry Ivanov, Evangelos Kranakis, Danny Krizanc, Oscar Morales-Ponce
SIROCCO 2025 — International Colloquium on Structural Information and Communication Complexity

- [17] **PISA: An Adversarial Approach To Comparing Task Graph Scheduling Algorithms**
Jared Coleman, Bhaskar Krishnamachari
IPDPS 2025 - The 39th International Parallel and Distributed Processing Symposium
- [16] **LLM-Assisted Rule Based Machine Translation for Low/No-Resource Languages**
Jared Coleman, Bhaskar Krishnamachari, Khalil Iskarous, Ruben Rosales
AmericasNLP @ NAACL 2024 - 4th Workshop on NLP for Indigenous Languages of the Americas
- [15] **Linear Search for an Escaping Target with Unknown Speed**
Jared Coleman, Dmitry Ivanov, Evangelos Kranakis, Danny Krizanc, Oscar Morales-Ponce
IWOCA 2024 — 35th International Workshop on Combinatorial Algorithms
- [14] **Online Allocation of Sensing and Computation in Large Graphs**
Xinlin Li, Merve Karakas, Osama A. Hanna, Mehrdad Kiamari, Jared Coleman, Christina Fragouli, Bhaskar Krishnamachari, Gunjan Verma
IEEE CIC 2023 — International Conference on Collaboration and Internet Computing
- [13] **DARSAN: A Decentralized Review System Suitable for NFT Marketplaces**
Sulyab Thottungal Valapu, Tamoghna Sarkar, Jared Ray Coleman, Anusha Avyukt, Hugo Embrechts, Dimitri Torfs, Michele Minelli, Bhaskar Krishnamachari
ICBC 2023 — International Conference on Blockchain
- [12] **Search and Rescue on the Line**
Jared Coleman, Lorand Cheng, Bhaskar Krishnamachari
SIROCCO 2023 — International Colloquium on Structural Information and Communication Complexity
- [11] **PhD Forum Abstract: Cooperative Problem-Solving with Systems of Constrained Mobile Agents**
Jared Ray Coleman
IPSN 2023 — International Conference on Information Processing in Sensor Networks
- [10] **Graph Convolutional Network-based Scheduler for Distributing Computation in the IoRT**
Jared Coleman, Mehrdad Kiamari, Lillian Clark, Daniel D’Souza, Bhaskar Krishnamachari
MILCOM 2023 WS-7 — Workshop On The Internet Of Things For Adversarial Environments
- [9] **Delivery to Safety with Two Cooperating Robots**
Jared Coleman, Evangelos Kranakis, Danny Krizanc, Oscar Morales-Ponce
SOFSEM 2023 — International Conference on Current Trends in Theory and Practice of Computer Science
- [8] **The Snow Plow Problem: Perpetual Maintenance by Mobile Agents on the Line**
Jared Coleman, Oscar Morales-Ponce
ICDCN 2023 — International Conference on Distributed Computing and Networking
- [7] **Line Search for an Oblivious Moving Target**
Jared Coleman, Evangelos Kranakis, Danny Krizanc, Oscar Morales-Ponce
OPODIS 2022 — International Conference on Principles of Distributed Systems
- [6] **Multi-Objective Network Synthesis for Dispersed Computing in Tactical Environments**
Jared Coleman, Eugenio Grippio, Bhaskar Krishnamachari, Gunjan Verma
SPiE Defense + Commercial Sensing 2022
- [5] **Network Synthesis for Tactical Environments: Scenario, Challenges, and Opportunities**
Tzanis Anevlavis, Jonathan Bunton, Jared Coleman, Mine Dogan, Eugenio Grippio, Abel Souza, Christina Fragouli, Bhaskar Krishnamachari, Matthew Maness, Karl Olson, Prashant Shenoy, Paulo Tabuada, Gunjan Verma
SPiE Defense + Commercial Sensing 2022
- [4] **Robotic Sorting on the Grid**
Jared Coleman, Oscar Morales-Ponce
ICDCN 2022 — 23rd International Conference on Distributed Computing and Networking
- [3] **Message Delivery in the Plane by Robots with Different Speeds**
Jared Coleman, Evangelos Kranakis, Oscar Morales-Ponce, Danny Krizanc
SSS 2021 — 23rd International Symposium on Stabilization, Safety, and Security of Distributed Systems

[2] **The Pony Express Communication Problem**

Jared Coleman, Evangelos Kranakis, Oscar Morales-Ponce, Danny Krizanc
IWOCA 2021 — 32nd International Workshop on Combinatorial Algorithms

[1] **Minimizing The Maximum Distance Traveled To Form Patterns With Systems of Mobile Robots**

Jared Coleman, Evangelos Kranakis, Oscar Morales-Ponce, Jorge Urrutia, Birgit Vogtenhuber
CCCG 2020, 32nd Canadian Conference on Computational Geometry, August 5-7, 2020

OTHER RESEARCH PRODUCTS

Kubishi Sentences (sentences.kubishi.com)

LLM-powered sentence building and translation tools for the Owens Valley Paiute Language 2024 - present

Kubishi Dictionary (dictionary.kubishi.com)

An online dictionary for the Owens Valley Paiute language 2020 - present

ChatLang (github.com/ANRGUSC/chatlang)

An LLM-powered role-playing bot with parallel tutor assistance 2023

GCN-Turtlebot (github.com/ANRGUSC/gcnschedule-turtlenet)

GCN-based Scheduler for Distributing Computation in the Internet of Robotic Things
First-place winner: 2nd Student Design Competition on Networked Computing on the Edge at CPS IoT Week 2022

Secure IIoT Application

Using blockchain technology to secure industrial IoT systems Industry Partner: Chevron 2021-2023

Aerocube @ The Beach

Distributed robotics systems for space — a proof of concept Industry Partner: The Aerospace Corporation 2017

MEDIA COVERAGE

[This new tech could save old languages from dying out](#)

Clare Duffy, CNN
September 2, 2025

[Can A.I. Help Revitalize Indigenous Languages?](#)

Serena Jampel, Smithsonian Magazine
July 31, 2025

[Why some endangered language speakers have to get creative with AI preservation efforts](#)

Nichole Currie, WHYY (PBS/NPR)
December 9, 2024

[Revitalizing Critically Endangered Languages via Large Language Models](#)

Loyola Marymount University Newsroom
November 14, 2024

[Imagine Hearing A Distant Relative Telling Stories in a Nearly Forgotten Language. What Would You Do?](#)

Greg Hardesty, USC Viterbi, School of Engineering
June 20, 2024